

Homework 5 Solutions

Based on the third edition of *Abstract Algebra* of Dummit and Foote.

Section 3.2: More on Cosets and Lagrange's Theorem

Section 2, Exercise 4

Show that if $|G| = pq$ for some primes p and q (not necessarily distinct) then either G is abelian or $Z(G) = 1$. [See Exercise 36 in Section 1.]

Solution. We prove the contrapositive of the nontrivial direction: assume $Z(G) \neq 1$ and show that G is abelian. By Lagrange's Theorem, $|Z(G)|$ divides pq . If $Z(G) = G$, there is nothing to prove. Otherwise, $|Z(G)|$ is one of the prime factors p or q (with the same conclusion when $p = q$), and hence

$$|G/Z(G)| = |G : Z(G)|$$

is prime. A group of prime order is cyclic, so $G/Z(G)$ is cyclic. The assertion follows from [Section 3.1, Exercise 36] (solved in [Homework 4](#)). $\square$

Section 2, Exercise 10

Suppose H and K are subgroups of finite index in the (possibly infinite) group G with $|G : H| = m$ and $|G : K| = n$. Prove that

$$\text{lcm}(m, n) \leq |G : H \cap K| \leq mn.$$

Deduce that if m and n are relatively prime then $|G : H \cap K| = |G : H| \cdot |G : K|$.

Solution. Put $L = H \cap K$. Consider the map of sets of left cosets

$$\Phi : G/L \longrightarrow G/H \times G/K, \quad gL \longmapsto (gH, gK).$$

This map is well defined: if $g_1L = g_2L$, then $g_2^{-1}g_1 \in L \subseteq H, K$, so both corresponding cosets agree. It is also injective. Indeed, if $g_1H = g_2H$ and $g_1K = g_2K$, then $g_2^{-1}g_1 \in H \cap K = L$, whence $g_1L = g_2L$. Therefore

$$|G : L| \leq |G/H| |G/K| = mn.$$

In particular, L has finite index in G .

Since $L \leq H \leq G$ and $L \leq K \leq G$, we have

$$|G : L| = |G : H| |H : L| = m |H : L| \text{ and } |G : L| = |G : K| |K : L| = n |K : L|.$$

Thus both m and n divide $|G : L|$, so their least common multiple divides this index. Hence

$$\text{lcm}(m, n) \leq |G : H \cap K| \leq mn.$$

If $(m, n) = 1$, then $\text{lcm}(m, n) = mn$. The two bounds coincide, and the assertion follows. $\square$

Section 2, Exercise 18

Let G be a finite group, let H be a subgroup of G and let $N \trianglelefteq G$. Prove that if $|H|$ and $|G : N|$ are relatively prime then $H \leq N$.

Solution. Let $\pi : G \rightarrow G/N$ be the natural projection, and restrict it to H . The First Isomorphism Theorem gives

$$\pi(H) \cong H/(H \cap N).$$

It follows that $|\pi(H)|$ divides $|H|$. On the other hand, $\pi(H)$ is a subgroup of G/N , so Lagrange's Theorem shows that $|\pi(H)|$ also divides

$$|G/N| = |G : N|.$$

The two displayed integers are relatively prime by hypothesis. Hence $|\pi(H)| = 1$, which means that every element of H maps to N . Therefore $H \leq N$. $\square$

Section 2, Exercise 19

Prove that if N is a normal subgroup of the finite group G and $(|N|, |G : N|) = 1$ then N is the unique subgroup of G of order $|N|$.

Solution. Let H be any subgroup of G with $|H| = |N|$. The hypothesis implies

$$(|H|, |G : N|) = (|N|, |G : N|) = 1.$$

Exercise 18 therefore applies and gives $H \leq N$. Since H and N are finite and have the same order, we get $H = N$. Thus every subgroup of G having order $|N|$ is N , so N is unique. $\square$

Section 3.3: The Isomorphism Theorems

Section 3, Exercise 3

Prove that if H is a normal subgroup of G of prime index p then for all $K \leq G$ either

- (i) $K \leq H$ or
- (ii) $G = HK$ and $|K : K \cap H| = p$.

Solution. Fix $K \leq G$. If $K \leq H$, (i) holds. Suppose instead that $K \not\leq H$. Because $H \trianglelefteq G$, the product HK is a subgroup of G . Moreover,

$$H < HK \leq G.$$

The quotient G/H has prime order p , whence $HK = G$.

Apply the Second Isomorphism Theorem to K and the normal subgroup H :

$$K/(K \cap H) \cong HK/H = G/H.$$

Taking orders gives $|K : K \cap H| = |G/H| = p$ whence (ii) holds. $\square$

1 Section 3.4: Composition Series and the Hölder Program

Section 4, Exercise 1

Prove that if G is an abelian simple group then $G \cong \mathbb{Z}_p$ for some prime p (do not assume G is a finite group).

Solution. Choose a nonidentity element $x \in G$. Because G is abelian, every subgroup of G is normal; in particular, the nontrivial subgroup $\langle x \rangle$ is normal. Simplicity therefore forces

$$G = \langle x \rangle,$$

so G is cyclic.

The generator x cannot have infinite order since that would force $G \cong \mathbb{Z}$. Hence x has finite order, say n .

The integer n must be prime. Indeed, if $n = ab$ with $a, b > 1$, then $\langle x^a \rangle$ has order b and is a nontrivial proper (hence normal) subgroup of G , again contradicting simplicity. Thus $n = p$ for some prime p . $\square$